

Graph Girth

Time limit: 1.00 s

Memory limit: 512 MB

Given an undirected graph, your task is to determine its *girth*, i.e., the length of its shortest cycle.

Input

The first input line has two integers n and m : the number of nodes and edges. The nodes are numbered $1, 2, \dots, n$.

After this, there are m lines describing the edges. Each line has two integers a and b : there is an edge between nodes a and b .

You may assume that there is at most one edge between each two nodes.

Output

Print one integer: the girth of the graph. If there are no cycles, print -1 .

Constraints

- $1 \leq n \leq 2500$
- $1 \leq m \leq 5000$

Example

Input	Output
5 6 1 2 1 3 2 4 2 5 3 4 4 5	3